Assignment

Q1. A belt transmission dynamometer, the driving pulley rotates at 300 rpm. The distance between the centre of the driving pulley and dead mass is 800 mm. The diameter of each of the driving as well as the intermediate pulley is equal to 360 mm. Find the value of the dead mass required to maintain the lever in a horizontal position when the power transmitted is 3000 watts. Also, find its value when the belt is just begins to slip on the driving pulley; μ being 0.25 and the maximum tension in the belt is 1200 N.

Q2. A vehicle having a wheel base of 3.2 m has its centre of mass at 1.4 m from the rear wheels and 55 mm from the ground level. It moves on a level ground at a speed of 54 km/hr. Determine the distance moved by the car before coming to rest on applying the brakes to the

(i) Rear wheels
(ii) Front wheels
(iii) All the four wheels.

The coefficient of friction between the ~~tyre~~ Tyres and the road is 0.5.

Q3. A band and block brake has 14 blocks. Each block subtends an angle of 14° at the centre of the rotating drum. The diameter of the drum is 750 mm and the thickness of the blocks is 65 mm. The two ends of the band are fixed to the pins on the lever at distances of 50 mm and 210 mm from the fulcrum on the opposite sides.
Determine the least force required to be applied

at the lever at a distance of 600 mm from the fulcrum if the power absorbed by the blocks is 180 kW at 175 rpm. Coefficient of friction between the blocks and the drum is 0·35.

A band and block brake having 12 blocks, each of which subtends an angle of 16° at the centre, is applied to a rotating drum with a diameter of 600 mm. The blocks are 75 mm thick. The drum and the flywheel mounted on the same shaft have a mass of 1800 kg and have a combined radius of gyration of 600 mm. The two ends of the band are attached to pins on the opposite sides of the brake fulcrum at distances of 40 mm and 150 mm from it. If a force of 0·25 kN is applied on the lever at a distance of 900 mm from the fulcrum, find the

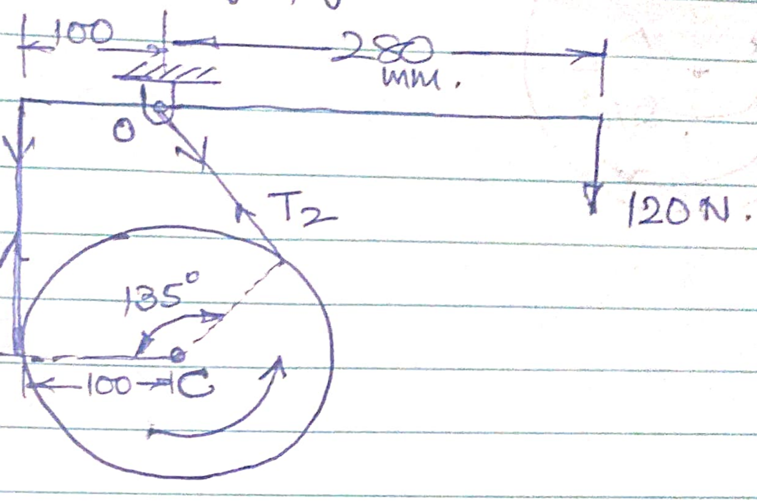
(a) Maximum braking torque
(b) Angular retardation of the drum
(c) Time taken by the system to be stationary from the rated speed of 300 rpm.

Take coefficient of friction between the blocks and the drum as 0·3.

A simple band brake is applied to a shaft carrying a flywheel of 250 kg mass and of radius of gyration of 300 mm. The speed of shaft is 200 rpm. The drum diameter is 200 mm and the coefficient of friction is 0·25. Determine

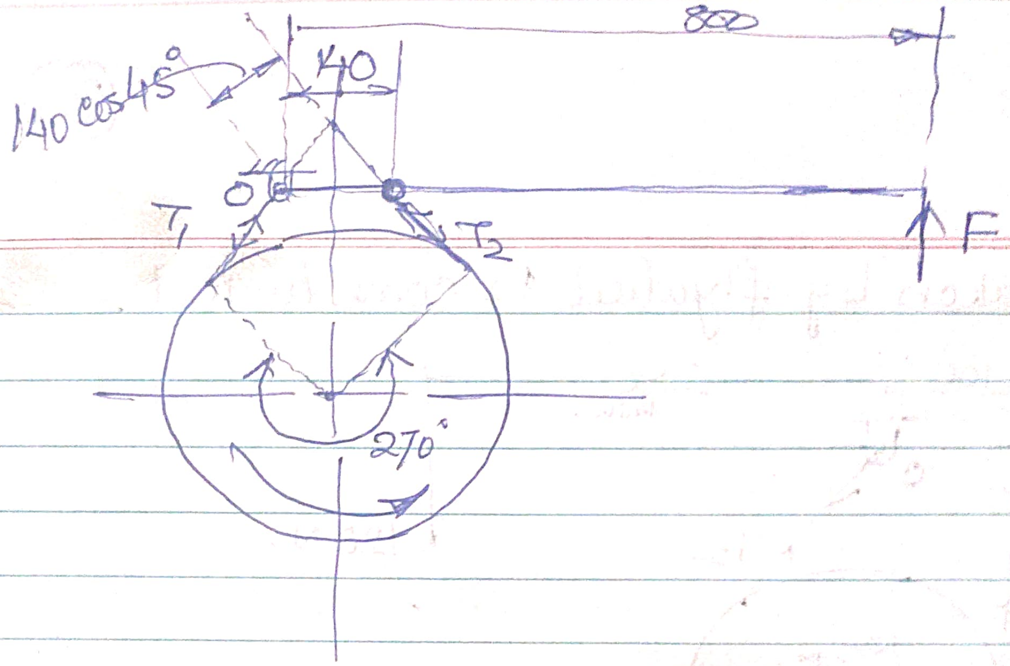
(a) Brake torque when a force of 120 N is applied at the lever end.
(b) Number of turns of the flywheel before it comes to rest.

(c) Time taken by flywheel to come to rest.



Q6. A differential band brake has a drum with a diameter of 800 mm. The two ends of the band are fixed to pins ~~to~~ on the opposite sides of the fulcrum of the lever at distances of 40 mm and 200 mm from the fulcrum. The angle of contact is 270° and the coeff. of friction is 0.2. Determine the brake torque when a force of 600 N is applied to the lever at distance of 800 mm from the fulcrum.

Q7. A simple band brake is applied to a drum of 560 mm diameter which rotates at 240 rpm. The angle of contact of the band is 270°. One end of the band is fastened to a fixed pin and the other end ~~of~~ to the brake lever, 140 mm from the fixed pin. The brake lever is 800 mm long and is placed perpendicular to the diameter that bisects the angle of contact. Assuming the coefficient of friction as 0.3, determine the necessary pull at the end of the lever to stop the drum if 40 kW of power being ~~used~~ absorbed. Also find the width of the band if its thickness is 3 mm. and maximum tensile stress is limited to 40 MPa.



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